

APPLICATIONS

SELECTION CRITERIA FOR REGRESSION
MODELS AND THEIR APPLICATION TO
ABNORMAL EVENT STUDIES IN FINANCE

- Graham D.I. Barr:** Department of Statistical Sciences, University of Cape Town, Private Bag, Rondebosch, Cape Town, 7701, Republic of South Africa.
- Ross Sparks:** Division of Mathematics and Statistics, CSIRO, P.O. Box 218, Linfield NSW2070, Australia.
- John Affleck-Graves:** College of Business Administration, University of Notre Dame, Notre Dame, Indiana 46556.
- Mary Lou Thompson:** Department of Statistics, University of Washington, WA98806.

Key words: Capital Asset Pricing Model; Event studies; linear regression; Market Model; Model Selection criteria; Zero intercept models.

Summary: This paper, which is concerned with the development of model selection criteria in the context of regression analysis, was motivated by the following application. The assessment of the impact of abnormal events, such as the announcement of a merger or financial restructuring, on share price performance presupposes the specification of a model that appropriately describes 'normal' share price behaviour. Traditionally, in the Finance literature, this model is taken to be a simple linear regression of excess share price returns on the excess return on the market index. In addition, two constrained forms of this simple model viz. a model with a zero intercept term and unconstrained market index coefficient (Capital Asset Pricing Model), and a model with a zero intercept term and coefficient of the market index that is put equal to 1 (Market adjusted model) have been considered as alternatives (Brown and Warner, 1980). An appropriate set of model selection criteria for evaluation of these models based on the MSEF (mean square error of

prediction) and the MSE (mean square error) principles are developed. The selection criteria are then applied to a large number of share prices on the New York and Johannesburg Stock Exchanges. In the case of the New York Stock Exchange, no particular model tended to be the dominant selection but in the case of the Johannesburg Stock Exchange the procedure was more unequivocal and in a very high percentage of cases a simple Market Adjusted Model rather than an unconstrained model was selected. This selection pattern is clearly related to the fact that, on average, share returns on the JSE exhibit lower fits against market returns than do share returns on the NYSE. This is related to the higher volatility of the JSE and the persistent problems of thin trading on this market. The analysis represents an interesting example of how unstable relationships are often more effectively described by fitting a model of lower complexity.

1. INTRODUCTION

Many empirical studies in Finance and Accounting are concerned with the impact of firm-specific events on the market value of the affected firm's securities. Investigations of this type are often termed "event studies" and include the examination of the impact of events such as earnings announcements (Ball and Brown 1968), stock splits (Fama et al., 1969), merger announcements (Asquith, 1983) and new equity offers by seasoned companies (Smith, 1977), as well as the effects of public announcements such as money supply announcements (Urich and Wachtel, 1981), and government regulation or deregulation (Schipper and Thompson, 1983) on security returns. A central feature of most event studies is the computation of abnormal returns, that is, the difference between *ex post*¹ return and some appropriate benchmark return. This benchmark return is usually obtained from an assumed market model which is estimated using data which excludes data points around and including the event. The estimated model is then used to compute abnormal returns over the period of the event.

¹ i.e. that which has already transpired

Two particular benchmark models that have been applied in the Finance and Accounting literature when computing abnormal returns are the mean adjusted model and the market model. The first, the mean adjusted model, computes abnormal returns as

$$AR_{it} = R_{it} - \bar{R}_i,$$

where

AR_{it} = the abnormal return on security i in period t ,

R_{it} = the observed return (ex post) on security i in period t ,

and

$\bar{R}_i$ = the average return on security i in the estimation period

(the range of data over which the analysis is being performed). The second, the market model, may be specified in many forms but all are of the following basic structure:

$$AR_{it} = R_{it} - (a_i + b_i R_{mt}),$$

where

R_{mt} = the return on the market (usually an index) in period t , and a_i , b_i are estimates of the intercept and slope parameters from the regression

$R_{it} = \alpha_i + \beta_i R_{mt} + \epsilon_{it}$ over the estimation period.

In particular, R_{it} and R_{mt} are usually expressed as returns in excess of the risk free rate (the rate of return on a financial contract with certain return, for example a bank deposit rate).

In two major simulation studies, Brown and Warner (1980, 1985) examined the power of the mean adjusted model and several forms of the market model to detect the presence of artificially introduced events in stock return data. Their results indicated that if event dates are clustered in time (fall within a narrow interval of time), the mean adjusted model can result in these tests having a significance level (size of the test) which is seriously misstated. Because many events in practice are clustered in time and

because there is no precise definition of what constitutes sufficient clustering to influence the size of the mean adjusted test, most researchers prefer to use the market model approach. Consequently this paper will focus on this approach. In particular, this paper will examine three distinct forms of the market model, the Capital Asset Pricing Model (CAPM), the Market Adjusted Model (MAM), and the Single Factor Model (SFM).

Given a single factor (market) return generating model, each of the above three models is appropriate under different assumptions². Thus, in practice, the researcher is faced with a choice between these three (or more) competing models. In addition, since the objective is to provide good predictions of the returns in the event period, the model should be selected on the basis of its prediction error. The problem of model selection for prediction purposes in a regression context has been widely considered in the statistical literature. Reviews are given by Hocking (1976) and Thompson (1978a, 1978b). More recently, Linhart and Zucchini (1986) have provided a broad methodological framework for approaching the problem of model selection. Their methodology covers the well-known C_p and S_p criteria for model selection in linear regression for fixed and random independent variables respectively.

We derive exact expressions for the appropriate model selection criteria for CAPM and MAM against the general single factor model. The CAPM result is a special case of the C_p criterion.

More specifically, we will develop precise criteria which allow the researcher to objectively choose between the CAPM, MAM and SFM for each individual security in an event study so as to minimize the mean square prediction error. The framework established can then be easily extended to other competing models. Note that there may be many cases where the model selection methodology does not significantly alter the results obtained

² for example, the SFM has applicability when security risk is assumed not to be influenced by market behaviour; the MAM has applicability when the systematic component of security risk is assumed to be constant, and the CAPM has applicability under more general circumstances (see Brown and Warner, 1980).

using just a single model such as the SFM for all stocks. However, as the researcher does not know a priori whether or not the results are sensitive to model selection, the method provides a flexible and easily implemented procedure which ensures that for each stock, on average, the most appropriate model is used.

Finally, we will provide some empirical results using actual stock market data which will indicate that neither of the above three models dominate completely but that in certain financial environments the simpler models appear superior.

The remainder of the paper is organized as follows. In section 2 we define a basic return generating model and the two approximating market models to be examined in this paper. In section 3 we develop the model selection criteria to be used in choosing between these three competing models on the basis of the mean square error of prediction (MSEP) criterion. Section 4 then provides some simple extensions to a mean square error (MSE) criterion which may generally be more appropriate if parameter estimation rather than prediction is the major focus of attention. Some empirical results are then presented in section 5 and the paper closes with a section of conclusions.

2. ALTERNATIVE MARKET MODELS

Single factor market models of stock returns can all be expressed as versions of the single factor model (SFM),

$$(1) \quad y_{it} = \alpha_i + \beta_i x_t + e_{it},$$

where:

y_{it} is the excess return on security i (i.e., $R_{it} - R_{ft}$),

x_t is the excess return on the market,

R_{ft} is the risk free rate,

e_{it} are independently and identically distributed over time (t),

$E(e_{it}) = 0$, and

$Var(e_{it}) = \sigma^2$.

While (1) is a general single factor return generating model, financial theory leads to consideration of two more specific models. These are the Sharpe-Lintner Capital Asset Pricing Model (CAPM) defined as

$$(2) \quad y_{it} = \beta_c x_t + u_{it},$$

where,

u_{it} are independently and identically distributed over time (t),

$$E(u_{it}) = 0, \text{ and}$$

$$\text{Var}(u_{it}) = \sigma_u^2,$$

and the Market Adjusted Model (MAM), defined as

$$(3) \quad y_{it} = x_t + v_{it}.$$

where,

v_{it} are independently and identically distributed over time (t),

$$E(v_{it}) = 0, \text{ and}$$

$$\text{Var}(v_{it}) = \sigma_v^2.$$

Clearly, the latter two merely impose assumptions on the values of the parameters α and β in the general model (1). Thus, the CAPM imposes the restriction that $\alpha = 0$, while the Market Adjusted Model imposes the restriction $\alpha = 0, \beta = 1$.

3. MODEL SELECTION

The issue at hand in this paper is whether in the general model (1) it is useful to make the simplifying assumptions implied by (2) or (3). The introduction of a fixed parameter value clearly introduces bias, unless the fixed value is constant across all securities and is equal to the selected value. However, it removes any variance of estimation from that parameter and this can result in smaller prediction errors.

In order to consider whether there is any net gain from the introduction of fixed value parameters one must first consider whether the main purpose of the estimation is parameter determination or the prediction of the dependent variable. One must then decide how best to measure the lack of fit of the

restricted model from some more general alternative. One framework within which to consider these issues is the model selection framework suggested by Linhart and Zucchini (1986). In this approach an operating model is specified which is a close representation of the true situation. This model is then compared to various families of approximating models by specifying a discrepancy which measures the lack of fit of the approximation. The approximating family which minimizes an estimate of the expected discrepancy is then selected. In our context, model (1) is the operating model while (2) and (3) are approximating models. For any particular data set the operating model and the approximating models have to be estimated. The discrepancy between any fitted model and the assumed operating model is called the overall discrepancy of that model. In the case of these approximating models, it comprises a bias term (the "discrepancy due to approximation") which is not data dependent, and a variance term (the "discrepancy due to estimation") which is dependent on the data and estimation method. The operating model itself can be estimated without bias (by assumption) but is subject to estimation error.

We will make the assumption that x_i is fixed and consider a particular security (thus dropping the i subscript). Using the approach of Linhart and Zucchini (1986) we define the operating model F to be the Market model

$$y_t = \alpha_0 + \beta_0 x_t + e_t,$$

where

$$e_t = y_t - E_F y_t, \quad t = 1, \dots, T$$

are independently and identically distributed with constant variance σ^2 .

We let $\{(y_t, x_t), t = 1, \dots, T\}$ be a "training sample" to fit this operating model. In this application we will consider two approximating models from the family G of linear approximating models with one independent variable.

Thus we have,

Operating model:

$$F: y_t = \alpha_0 + \beta_0 x_t + e_t \quad (\text{Market Model})$$

Approximating Models:

$$G_1 : y_t = \beta_c x_t + u_t \quad (\text{CAPM})$$

$$G_2 : y_t = x_t + v_t \quad (\text{Market Adjusted Model})$$

The overall discrepancy we use for a particular model, M , is the total mean square error of prediction (MSEP) taken over the training sample. On the premise that the x -values in the training sample are the region of interest, $\hat{y}_{M,t}$ denotes the estimate of the expected value of y based on the training sample. The MSEP is the expected squared distance of subsequent realisations $\tilde{y}_t$ (corresponding to subsequent x_t) from these fitted values, then expressed as:

$$\Delta(\hat{M}, F) = \sum_{t=1}^T E_F(\tilde{y}_t - \hat{y}_{M,t})^2$$

This is a measure of the total distance between subsequent y 's and the model M estimated from the training sample. Being dependent on the training sample, it is a random variable.

Note that M may be F in the case of the operating model or G_1 or G_2 in the case of the approximating models and the expectations are taken over F , but that the training sample is taken as fixed. The $\tilde{y}_t$ are regarded as subsequent realisations corresponding to each of $x_1, x_2, \dots, x_T$ as in the training sample and $\hat{y}_{M,t}$ denotes the corresponding estimate of $E(y_t)$ under M , from the (fixed) training sample using Ordinary Least Squares (OLS) to estimate the parameters of the model M .

The expected overall discrepancy for M (averaged over training samples) is then defined as

$$E(\Delta(\hat{M}, F)) = \sum_{t=1}^T E_{F^*}(\tilde{y}_t - \hat{y}_{M,t})^2$$

where the expectations are now taken over F and $\hat{y}_{M,t}$, $t = 1, 2, \dots, T$,

notated (by Linhart and Zucchini, 1986) as F^* . We may thus consider the calculation of estimates of the expected overall discrepancy (referred to by Linhart and Zucchini (1986) simply as the expected discrepancy) using the two approximating models mentioned above and may thus make comparisons between competing models.

As mentioned above, the primary focus of this paper is to test the applicability of certain simple financial models against a more generally used but more complicated alternative. These models could be gauged against the alternative by considering whether they produced superior predictions for the current data (MSEP approach), or superior parameter estimates for the data set (MSE approach). The appropriate comparison will thus be between the expected overall discrepancy for the operating model and that for one of the approximating models. In order to estimate these quantities we develop *criteria* below in which the expected overall discrepancies can be estimated from observed data sets which would correspond to those observed in practice.

The first case we consider is that where we estimate the expected overall discrepancy of the operating model itself. As already mentioned, this model has zero discrepancy due to estimation but a non-zero discrepancy due to estimation.

The Operating Model: (Market Model) $y_t = \alpha_0 + \beta_0 x_t + e_t$

In this case we are using the operating model and thus $\hat{y}$ is an unbiased estimate of y . Thus,

$$\begin{aligned} E(\Delta(\hat{F}, F)) &= \sum_{t=1}^T E_{F^*}(\tilde{y}_t - \hat{y}_{Ft})^2 \\ &= \sum_{t=1}^T E_{F^*}(\tilde{y}_t - (\hat{\alpha}_0 + \hat{\beta}_0 x_t))^2 \end{aligned}$$

This is a special case ($p = k = 2$) of the result given by Thompson (1978b), page 133, from which it can be seen that:

$$E(\Delta(\hat{F}, F)) = T\sigma^2 + \sum_{t=1}^T \text{Var}(\hat{\alpha}_0 + \hat{\beta}_0 x_t) = (T+2)\sigma^2$$

Note that the expected overall discrepancy for the operating model consists entirely of discrepancy due to estimation as there is no discrepancy due to approximation because we have assumed the Market Model is the true model.

The Approximating Models:

G_1 Case: (CAPM) $y_t = \beta_c x_t + u_t$

Here,

$$\begin{aligned} (4) \quad E(\Delta(\hat{G}_1, F)) &= \sum_{t=1}^T E_{F^*}(\tilde{y}_t - \hat{y}_{G_1,t})^2 \\ &= T\sigma^2 + \sum_{t=1}^T \text{Var}(\hat{\beta}_c x_t) + \sum_{t=1}^T (E_{F^*}(\hat{\beta}_c x_t) - (\alpha_0 + \beta_0 x_t))^2 \\ &= (T+1)\sigma^2 + T\hat{\alpha}_0^2(1 + \hat{\theta}_x^2)^{-1} \end{aligned}$$

with $\hat{\theta}_x^2 = \frac{\bar{x}}{S_x^2}$, where $\bar{x}$ is the sample mean of the x_t and

$$S_x^2 = \sum_t \frac{(x_t - \bar{x})^2}{T},$$

is the maximum likelihood estimate of σ_x^2 .

In the parlance of Linhart and Zucchini (1986), this expected overall discrepancy has to be estimated to obtain a *criterion*. We would normally compute an unbiased estimate of σ^2 using the residuals of the operating model to obtain

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^T (y_t - (\hat{\alpha}_0 + \hat{\beta}_0 x_t))^2}{T-2}.$$

The criterion is then

$$(T+1)\hat{\sigma}^2 + T\hat{\sigma}_0^2(1 + \hat{\theta}_x^2)^{-1}.$$

The C_p criterion (Mallows, 1973) was developed as a scale invariant criterion. This allows comparison of discrepancies across different data sets. A scale invariant expression for testing the SFM operating model against the CAPM would be:

$$\frac{E(\Delta(\hat{G}_1, F)) - E(\Delta(\hat{F}, F))}{\sigma^2} = \frac{T\hat{\alpha}_0^2(1 + \hat{\theta}_z^2)^{-1}}{\sigma^2} - 1.$$

If σ^2 were known, an unbiased estimator of this difference is

$$\frac{T\hat{\alpha}_0^2(1 + \hat{\theta}_z^2)^{-1}}{\sigma^2} - 2.$$

σ^2 is, however, in general unknown. An unbiased estimator of the above, which acts as a criterion, is then

$$C_1 = \frac{T(T - 4)\hat{\alpha}_0^2(1 + \hat{\theta}_z^2)^{-1}}{(T - 2)\hat{\sigma}^2} - 2.$$

C_1 may be used to test the CAPM against the SFM. If $C_1 > 0$ it implies that the estimated overall discrepancy of the SFM is smaller than the estimated overall discrepancy of the CAPM and that the SFM is the superior model. If $C_1 < 0$ the reverse applies.

G_2 case: Market Adjusted Model, $y_t = x_t + v_t$

Here,

$$\begin{aligned} (5) \quad E(\Delta(\hat{G}_2, F)) &= \sum_{t=1}^T E_{F^*}(\tilde{y}_t - \hat{y}_{G_2, t})^2 \\ &= \sum_{t=1}^T E_{F^*}(\tilde{y}_t - x_t)^2 \\ &= T\alpha_0^2 + 2\alpha_0(\beta_0 - 1) \sum_{t=1}^T x_t + (\beta_0 - 1)^2 \sum_{t=1}^T x_t^2 + T\sigma^2. \end{aligned}$$

A criterion for this expression can be shown to be

$$T\hat{\alpha}_0^2 + 2\hat{\alpha}_0(\hat{\beta}_0 - 1)\sum_{t=1}^T x_t + (\hat{\beta}_0 - 1)^2 \sum_{t=1}^T x_t^2 + T\hat{\sigma}^2.$$

A scale invariant criterion for testing the MAM operating model against the SFM would be:

$$\frac{E(\Delta(\hat{G}_2, F) - E(\Delta(\hat{F}, F)))}{\sigma^2} = \frac{T\alpha_0^2 + 2\alpha_0(\beta_0 - 1) \sum_{t=1}^T x_t + (\beta_0 - 1)^2 \sum_{t=1}^T x_t^2}{\sigma^2} - 2.$$

An unbiased estimator of this expression in the general case where σ^2 is unknown would be

$$C_2 = \left[\frac{T-4}{T-2} \right] \frac{(T\hat{\alpha}_0^2 + 2\hat{\alpha}_0 - 1)\sum_{t=1}^T x_t - (\hat{\beta}_0 - 1)^2 \sum_{t=1}^T x_t^2}{\hat{\sigma}^2} - 4.$$

C_2 may be used as a criterion to test the MAM against the SFM.

If $C_2 > 0$ it implies that the estimated overall discrepancy of the SFM is smaller than the estimated overall discrepancy of the MAM and that the SFM is the superior model. If $C_2 < 0$ the reverse applies.

Note that the above overall expected discrepancy results (4) and (5) can also be obtained as special cases of a general result for the MSEP of a linear model using the following generalised approach:

In the linear model: $\underline{Y} = \underline{\mu} + \underline{e}$ where $\underline{\mu} : T \times 1$ belongs to some linear space $M \subset \mathbb{R}^T$, and where $E(\underline{e}) = \underline{0}$; $Cov(\underline{e}) = \sigma^2 I_T$. Then, if we consider the possibility that $\underline{\mu} \in L$, where L is some subspace of M , the least squares estimator of $\underline{\mu}$ is merely $P_L \underline{Y}$, the projection of $\underline{Y}$ onto L and the MSEP associated with the model L is:

$$E \| P_L \underline{Y} - \underline{Y}^* \|^2 = [T + \dim(L)]\sigma^2 + \| \underline{\mu} - P_L \underline{\mu} \|^2,$$

where $\underline{Y}^*$ is some future copy of $\underline{Y}$, $\dim(L)$ is the dimension of L and $\| \underline{a} \|^2 = \sum a_i^2$ is the squared norm of the vector $\underline{a}$.

Then, for the SFM, L is the 2-dimensional subspace spanned by the two vectors $\underline{1}$ and $\underline{x}_i$ for the CAPM, L is the 1-dimensional space spanned by $\underline{x}_i$ and for the MAM, L is completely pre-specified and has 0 dimension. In each case, $\|\underline{\mu} - P_{L\mu}\|^2$ represents the difference between the operating model and the approximating model.

Model selection:

Both C_1 and C_2 have been derived under the assumption that the operating model, $y_i = \alpha_0 + \beta_0 x_i + e_i$, is the true model. They may therefore be used to test the hypothesis that the approximating model produces better predictions than the operating model at the 50% level (see Linhart and Zucchini, 1986, p. 15). C_1 and C_2 thus have an interpretation as a test statistic of the various approximating models against the true model when the true model holds. Thus when $C_1 < 0$ one can select the CAPM approximating model as a superior prediction model to the SFM (operating) model, and when $C_2 < 0$ one can select the MAM approximating model as a superior prediction model to the SFM (operating) model.

C_1 and C_2 represent bilateral criteria. Quite clearly, if one were to do a 3-way comparison between SFM, CAPM and MAM one could simply compare the overall discrepancies.

C_1 and C_2 provide conditions under which, even when the operating model is the true model, the approximating models provide predictions with smaller MSE. Within a hypothesis testing framework, however, one generally assumes that *the simpler model is the correct one*. Following this route allows us to establish the familiar hypothesis tests from the scale invariant discrepancy estimators. Thus, if the CAPM were the true model, that is

with $H_0 : \alpha_0 = 0$,

and under the normality condition,

$$T\hat{\alpha}_0^2(1 + \hat{\theta}_x^2)^{-1}/\hat{\sigma}^2 \sim F_{1, T-2},$$

provides the appropriate test statistic. Under the alternative, the conditional

distribution is non-central F with non-centrality parameter equal to $T\alpha_o^2(1 + \hat{\theta}_m^2)^{-1}/\sigma^2$ and this can be used to compute the power of the hypothesis test.

Similarly, when the MAM is correct, that is

with $H_0: \alpha_o = 0$ and $\beta_o = 1$,

and under the normality condition,

$$\left[T\hat{\alpha}_0^2 + 2\hat{\alpha}_o(\hat{\beta}_o - 1) \sum_{t=1}^T x_t + (\hat{\beta}_o - 1)^2 \sum_{t=1}^T x_t^2 \right] / \hat{\sigma}^2 \sim 2F_{2, T-2},$$

provides the appropriate test statistic. Under the alternative, the conditional distribution is non-central F with non-centrality parameter equal to

$$\left[T\alpha_o^2 + 2\alpha_o(\beta_o - 1) \sum_{t=1}^T x_t + (\beta_o - 1)^2 \sum_{t=1}^T x_t^2 \right] / \sigma^2,$$

and this can be used to compute the power of the test.

4. EXTENSIONS TO MSE CASE

In the previous section, selection criteria C_1 and C_2 were developed in terms of the mean square error of prediction. This is the case of most interest in event studies. However, if the focus of interest is on the actual parameter estimates rather than on prediction outside the range of the data, the model with the most stable parameters should be selected. In these cases the emphasis for a selection procedure shifts from a MSEP to a MSE criterion.

The Mean Square Error (MSE) of an estimator $\hat{\gamma}$ for a parameter γ is defined as $E(\hat{\gamma} - \gamma)^2$. The MSE of a vector estimator $\hat{\gamma}$ for a vector of parameters γ with components γ_i , $i = 1, \dots, p$ may be defined as:

$$E \left[\sum_{i=1}^p (\hat{\gamma}_i - \gamma_i)^2 \right]$$

Thus the MSE for the class of models considered above would be of the form

$$E[(\hat{\alpha} - \alpha_0)^2 + (\hat{\beta} - \beta_0)^2]$$

where, for example, in the case of MAM: $\hat{\alpha} = 0$ and $\hat{\beta} = 1$.

In the case of the CAPM, it can be shown that the identical criterion (C_1) obtained under the MSEP procedure is obtained using the MSE procedure. For the MAM case the criterion obtained differs by a constant and can be shown to equal

$$C_2 + 1$$

Note that this MSE condition lies mid-way between that which is operative when the operating model is true and when the MAM approximating model is true using the MSEP criterion. This indicates that the MSE condition ($C_2 + 1$) is less likely to select MAM in preference to the operating model than under the MSEP procedure.

5. SOME EMPIRICAL RESULTS

Previous sections have derived criteria based on the MSEP and MSE respectively for choosing between a general single factor return generating model, the capital asset pricing model and the Market Adjusted Model respectively. To examine whether the two approximating models would ever be selected using actual stock return data the following empirical procedure was followed.

Two sets of data were considered. Daily data from the New York stock exchange, taken from July 1962 to December 1988 and daily data from the Johannesburg stock exchange from January 1973 to December 1989. In each case one company was chosen at random and a pseudo event date (day 0) was chosen at random from the price history for the stock. This was done with the proviso that at least one year's data be available for the stock and that the pseudo event date be chosen so that at least one year's data is available prior to the chosen pseudo date. Using the daily returns from 239 days prior to 10 days prior to the event day (this procedure was chosen to mimic the procedure adopted in many event studies using daily data and was also used by Brown and Warner, 1985, in their simulation study), the MSEP

criteria were applied for both the CAPM and the MAM. The procedure was then repeated for 50 000 randomly selected stocks (sampled with replacement) and the number of times the approximating model was selected over the operating model was counted. In addition, the entire procedure was repeated using estimation periods of 60 days (70 days prior to 10 days prior) and 20 days (30 days prior to 10 days prior) respectively and using monthly returns instead of daily returns (with an estimation period of 60 months prior to the "event" month). Note that for the monthly data we use the CRSP monthly file which has data from 1926 to 1988. The results are presented in table 1.

TABLE 1. Percentage of Time Approximating Model Selected Over the Operating Model in 50 000 Random Cases Using MSEP Criteria³

New York Stock Exchange				
Approximating Model	Daily Data		Monthly Data	
	T = 229	T = 60	T = 20	T = 60
CAPM	89.95%	90.43%	89.61%	81.62%
MAM: MSEP Criterion	48.02%	68.59%	78.47%	46.17%
MAM: MSE Criterion	41.04%	60.12%	70.63%	38.89%

Johannesburg Stock Exchange				
Approximating Model	Daily Data		Monthly Data	
	T = 229	T = 60	T = 20	T = 60
CAPM	96.24%	98.62%	98.42%	92.12%
MAM: MSEP Criterion	65.76%	80.27%	86.37%	64.21%
MAM: MSE Criterion	52.63%	72.34%	82.32%	49.77%

³ The operating model is $y_t = \alpha_o + \beta_o x_t + e_t$. The CAPM is the Sharpe-Lintner version of this model defined as $y_t = \beta_c x_t + u_t$. The market-adjusted model is $y_t = x_t + v_t$. y_t is the excess return on the stock in period t (i.e. the return on the stock less the risk-free rate), and x_t is the excess return on the market. T = the number of observations in the estimation period.

As can be seen from table 1, the CAPM is selected over the full model in more than 80% of the cases. The Market Adjusted Model was selected over the full model in just under 50% of the cases when using more common estimation periods (229 days for daily data and 60 months for monthly data). Also, as expected, the frequency with which the approximating model is selected over the full operating model does vary with T . For small T , the lack of precision in estimating the parameters in the full model results in the Market Adjusted Model being selected more frequently.

Although the CAPM appears better than the full operating model in over 80% of cases for the NYSE and in over 90% of cases for the JSE, this does not mean that it should be used for all stocks. The major purpose of this paper is to demonstrate that the model selection criteria in the statistics literature can be easily applied to each individual stock in an event study and the most appropriate model in terms of MSEP can then be used for each stock.

Finally, the above analysis is repeated using the MSE criterion. As noted in section 4, this produces identical results for the CAPM case. Thus, only the MAM results are presented in table 1 (third row). As expected, the approximating model (MAM) is selected less frequently using the MSE criterion than using the MSEP criterion (row 2 vs. row 3). However, even using this more restrictive criterion, our results indicate that selection of the MAM will result in lower MSE in over one third of the cases using conventional estimation periods.

In the case of the Johannesburg Stock Exchange data, it is noted that in all cases the approximating models are selected more frequently than in the cases using the CRSP data. Thin trading problems in particular have meant that the market model fits on the JSE data have been somewhat poorer than on major stock exchanges. The average R^2 (adjusted for degrees of freedom) for the full model estimation on the South African data was a low 0.161 whereas that for the New York data was 0.232.

Thus, although the method has advantages for all abnormal return

studies, it is seen as particularly useful for the smaller, more volatile markets which are often subject to thin trading and where the fits of share returns with respect to the market tend to be weaker.

6. SUMMARY AND CONCLUSIONS

The major purpose of this paper is to apply the Linhart and Zucchini (1986) framework to the derivation of model selection criteria that are applicable to event studies. Two commonly used approximating models are examined with reference to a general, one factor return generating model, and model selection criteria are derived based on unbiased estimates of a scale invariant relative overall discrepancy measure. Empirical results are presented using conventional estimation periods and actual stock return data which show that none of the three models examined will completely dominate all others. However, the procedure has been shown to be particularly useful in a market with thin trading and high volatility. Thus we provide a methodology which allows event study researchers to select the most appropriate model for each individual stock examined. Finally, we show how the MSEP criteria can be easily amended to the MSE criteria if estimation is the focus of the study.

7. ACKNOWLEDGEMENTS

Thanks to an anonymous referee for helpful suggestions, including pointing out to us the alternative route to the MSEP results.

REFERENCES

- ASQUITH, P. (1983). Merger bids, uncertainty, and stockholder returns. *Journal of Financial Economics*, 11, 51-83.
- BALL, R. & BROWN, P. (1968). An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6, 159-178.
- BROWN, S.J. & WARNER, J.B. (1980). Measuring security price information. *Journal of Financial Economics*, 8, 205-258.
- BROWN, S.J. & WARNER, J.B. (1985). Using daily stock returns: The case of event studies. *Journal of Financial Economics*, 14, 3-31.

- FAMA, E.F., FISHER, L., JENSEN, M.C. & ROLL, R. (1969). The adjustment of stock prices to new information. *International Economic Review*, 10, 1-21.
- HOCKING, R.R. (1976). The analysis and selection of variables in linear regression. *Biometrics*, 32, 1-49.
- LINHART, H. & ZUCCHINI, W. (1986). *Model selection*. John Wiley, New York.
- MALLOWS, C.L. (1973). Some comments on C_p , *Technometrics*, 15, 661-675.
- SCHIPPER, K. & THOMPSON (1983). The impact of merger-related regulations on the shareholders of acquiring firms. *Journal of Accounting Research*, 21, 184-221.
- SMITH, C.W. (1977). Alternative methods for raising capital: Rights versus underwritten offerings. *Journal of Financial Economics*, 5, 273-307.
- THOMPSON, M.L. (1978a). Selection of variables in multiple regression, Part I. *International Statistical Review*, 46, 1-19.
- THOMPSON, M.L. (1978b). Selection of variables in multiple regression, Part II. *International Statistical Review*, 46, 129-146.
- URICH, T. & WACHTEL, P. (1981). Market response to the weekly money supply announcements in the 1970s. *Journal of Finance*, 36, 1063-1072.